Lockheed SR-71 Blackbird

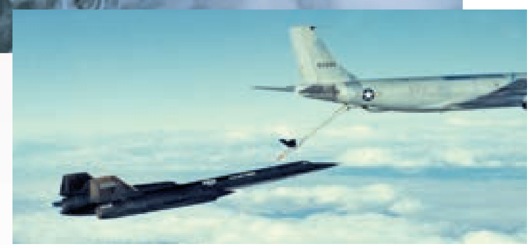
"I took off late on a winter afternoon, heading east where it was already dark... you streaked from bright day and flew into utter black."

Colonel Jim Wadkins
SR-71 pilot

A product of Lockheed's "Skunk Works" and chief designer Kelly Johnson, the SR-71 Blackbird may qualify as the most remarkable aircraft ever built. Developed in the early 1960s for the CIA with US Air Force dollars, under total secrecy, it involved radical innovation in almost every feature, from the materials used for the airframe and engine, through to the hydraulic system and the fuel. The result was the fastest jet-powered, crewed aircraft ever, capable of flying at altitudes of up to 100,000ft (30,000m) at more than three times the speed of sound.

When test pilots first saw the Blackbird prototypes at the CIA's Groom Lake site in Nevada in the early 1960s, they were taken aback by their shape and size – the elongated, slender fuselage and the huge engine pods mounted on the wings. The engines were actually wider than the main body of the fuselage. The radar-absorbent black paint, which gave the plane its name, covered a skin of titanium alloy, a lightweight, heat-resistant material that posed several problems for Lockheed engineers. It was too hard and brittle for most existing machine tools to work, and it was extraordinarily sensitive to contamination. Yet Johnson's instinct in persisting with titanium alloy proved justified. It resisted very high temperatures in Mach 3 flight, although the metal skin on the airplane's nose regularly wrinkled from the heat. The ground crew smoothed it out after a flight using a blowtorch – SR-71 pilot Colonel Jim Wadkins described the process as "like ironing a shirt."

The SR-71 entered service in 1966 and successfully performed the global reconnaissance role for which it was designed. Lockheed had ambitious plans for other versions of the aircraft, especially a high-altitude fighter interceptor prototyped as the YF-12A. The company claimed that 93 of these aircraft armed with air-to-air missiles would be able to defend the entire United States from attack by Soviet bombers, but the US government would not fund this costly project. The Blackbird was retired from service in 1999.



FILLING UP
Since oxygen is explosive when transported at the speeds attainable by the SR-71, the fuel tanks are purged with liquid nitrogen prior to filling (top). In-flight refueling (above) is used to extend the SR-71's already impressive range.

SHOCK DIAMONDS
As the SR-71B takes off, "shock diamonds" appear in the exhaust gases. These form when shock waves are reflected as they exit the engine's exhaust nozzles. The diamonds glow brightly as the surplus fuel ignites.